

BS – 101 Engineering Physics

Lecture – 15

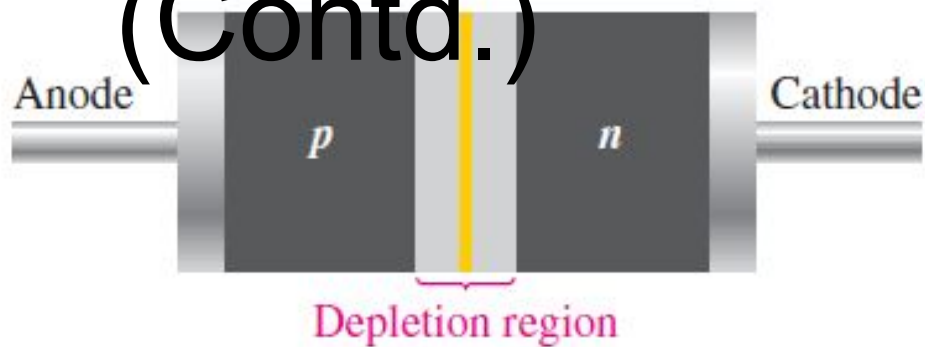
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Diod

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- **Diode** is made from a small piece of semiconductor material, usually silicon, in which half is doped as a p region and half is doped as an n region with a pn junction and depletion region in between.
- The p region is called the **anode** and is connected to a conductive terminal.
- The n region is called the **cathode** and is connected to a second conductive terminal.
- The basic diode structure and schematic symbol are shown in Figure a and b on next slide

Diode (Contd.)



(a) Basic structure



(b) Symbol

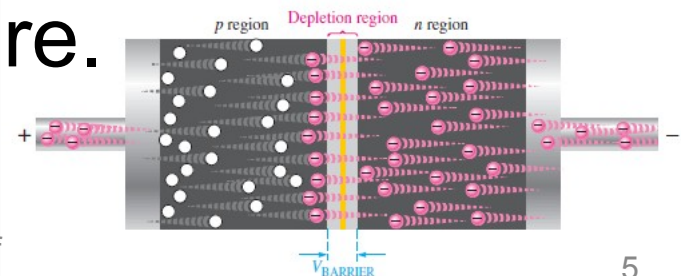


Diode (Contd.)

- Different types of diodes are mentioned in the reference material.
- Students are advised to look into it.

Forward Bias

- To bias a diode, we apply a dc voltage across it.
- Forward bias is a condition that allows current through pn junction.
- Battery Source is connected in such a way that positive side is connected to p region and negative side is connected to n region as shown in figure.



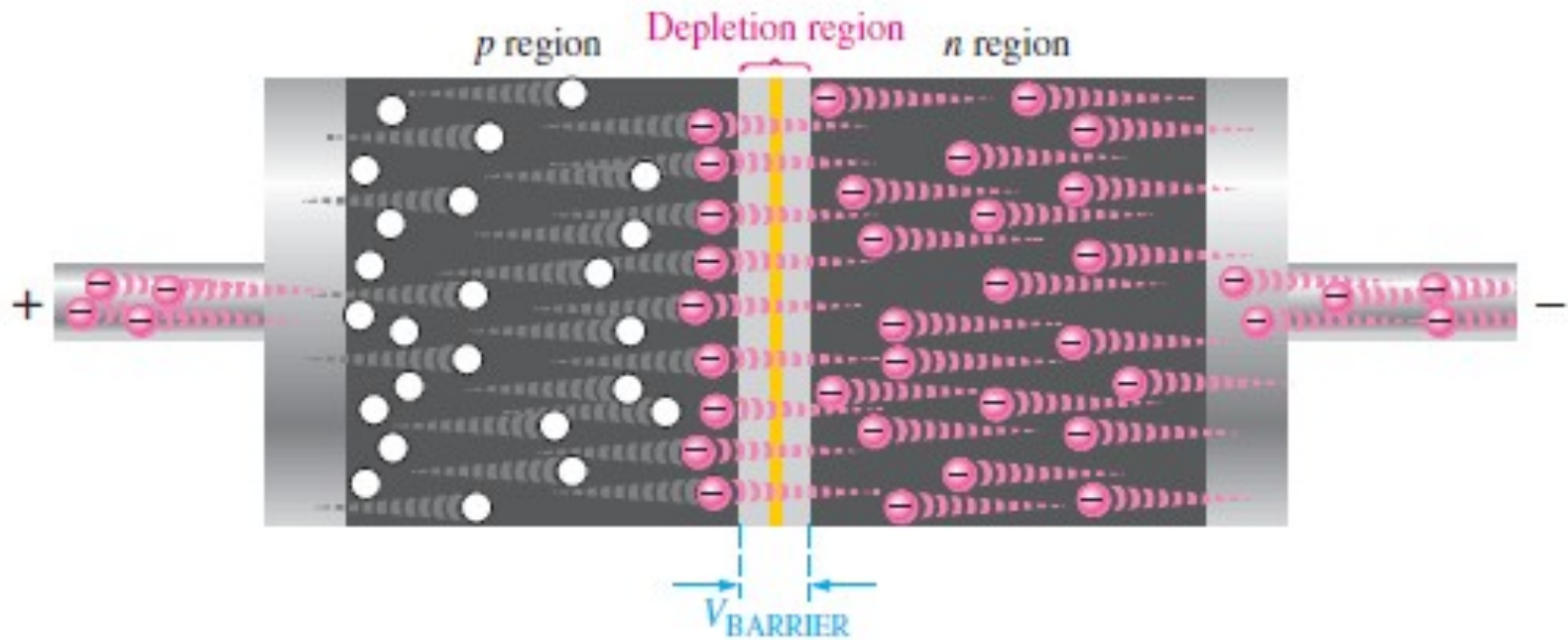
Forward Bias (Requirements)

- A dc voltage source connected across a diode in the direction to produce forward bias. This external bias voltage is designated as V_{BIAS} .
- Notice that the negative side V_{BIAS} is connected to the n region of the diode and the positive side is connected to the p region. This is one requirement for forward bias.
- A second requirement is that the bias voltage, V_{BIAS} , must be greater than the barrier

Forward Bias (Movement of Electrons)

- Now, the electrons are in the valence band in the p region, simply because they have lost too much energy overcoming the barrier potential to remain in the conduction band. Since unlike charges attract, the positive side of the bias-voltage source attracts the valence electrons toward the left end of the p region. The holes in the p region provide the medium or “pathway” for these valence electrons to move through the p region. The valence electrons move from one hole to the next toward the left. The holes, which are the majority carriers in the p region, effectively (not actually) move to the right toward the junction, as you can see in Figure. This *effective* flow of holes is the hole current. You can also view the hole current as being created by the flow of valence electrons through the p region, with the holes providing the only means for these electrons to flow.

Forward Bias (Movement of Electrons Contd.)





Forward Bias (Movement of Electrons Contd.)

- As the electrons flow out of the p region through the external connection (conductor) and to the positive side of the bias-voltage source, they leave holes behind in the p region; at the same time, these electrons become conduction electrons in the metal conductor. Recall that the conduction band in a conductor overlaps the valence band so that it takes much less energy for an electron to be a free electron in a conductor than in a semiconductor and that metallic conductors do not have holes in their structure. There is a continuous availability of holes effectively moving toward the pn junction to combine with the continuous stream of electrons as they come across the junction into the p region.



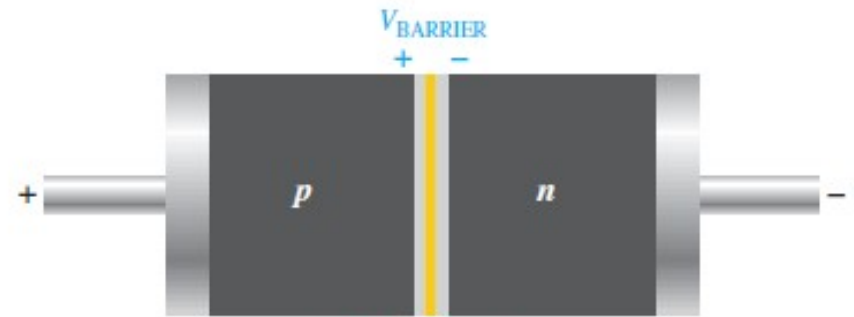
Effect of Forward Bias on Depletion Region

- As more electrons flow into the depletion region, the number of positive ions is reduced. As more holes effectively flow into the depletion region on the other side of the pn junction, the number of negative ions is reduced. This reduction in positive and negative ions during forward bias causes the depletion region to narrow, as indicated in Figure a & b.

Effect of Forward Bias on Depletion Region (Contd.)



(a) At equilibrium (no bias)



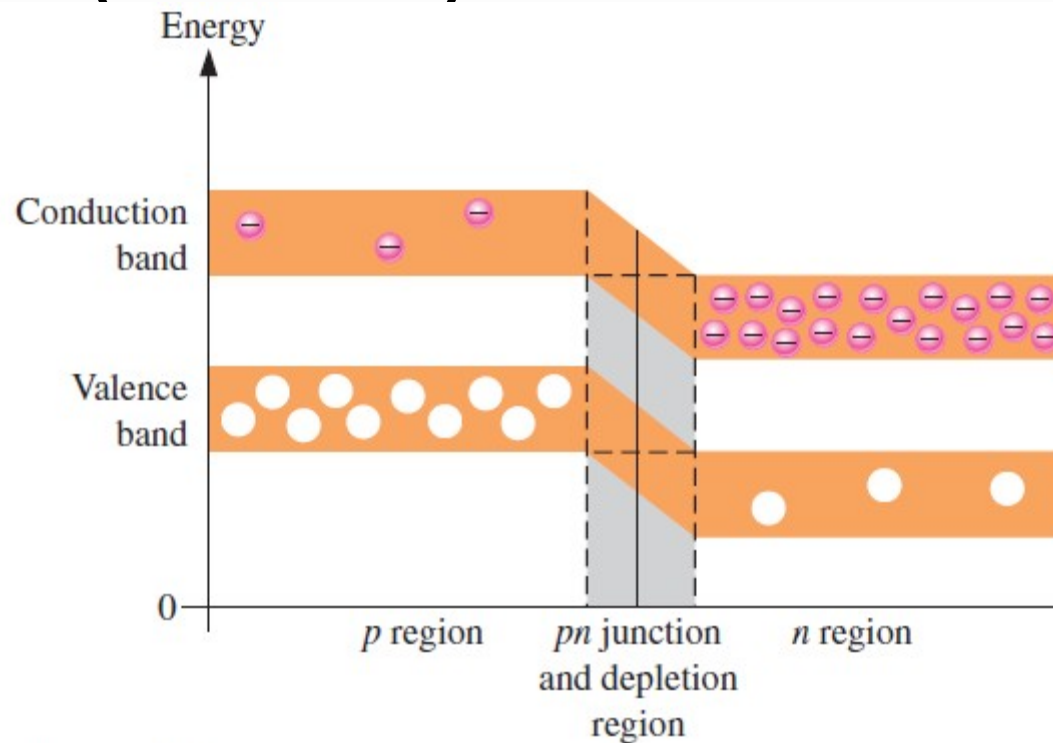
(b) Forward bias narrows the depletion region and produces a voltage drop across the pn junction equal to the barrier potential.



Effect of Barrier Potential During Forward Bias

- The electric field between the positive and negative ions in the depletion region on either side of the junction creates an “energy hill” that prevents free electrons from diffusing across the junction at equilibrium. This is known as the *barrier potential*.
- When forward bias is applied, the free electrons are provided with enough energy from the bias-voltage source to overcome the barrier potential and effectively “climb the energy hill” and cross the depletion region.
- The energy that the electrons require in order to pass through the depletion region is equal to the barrier potential.

Effect of Barrier Potential During Forward Bias (Contd.)



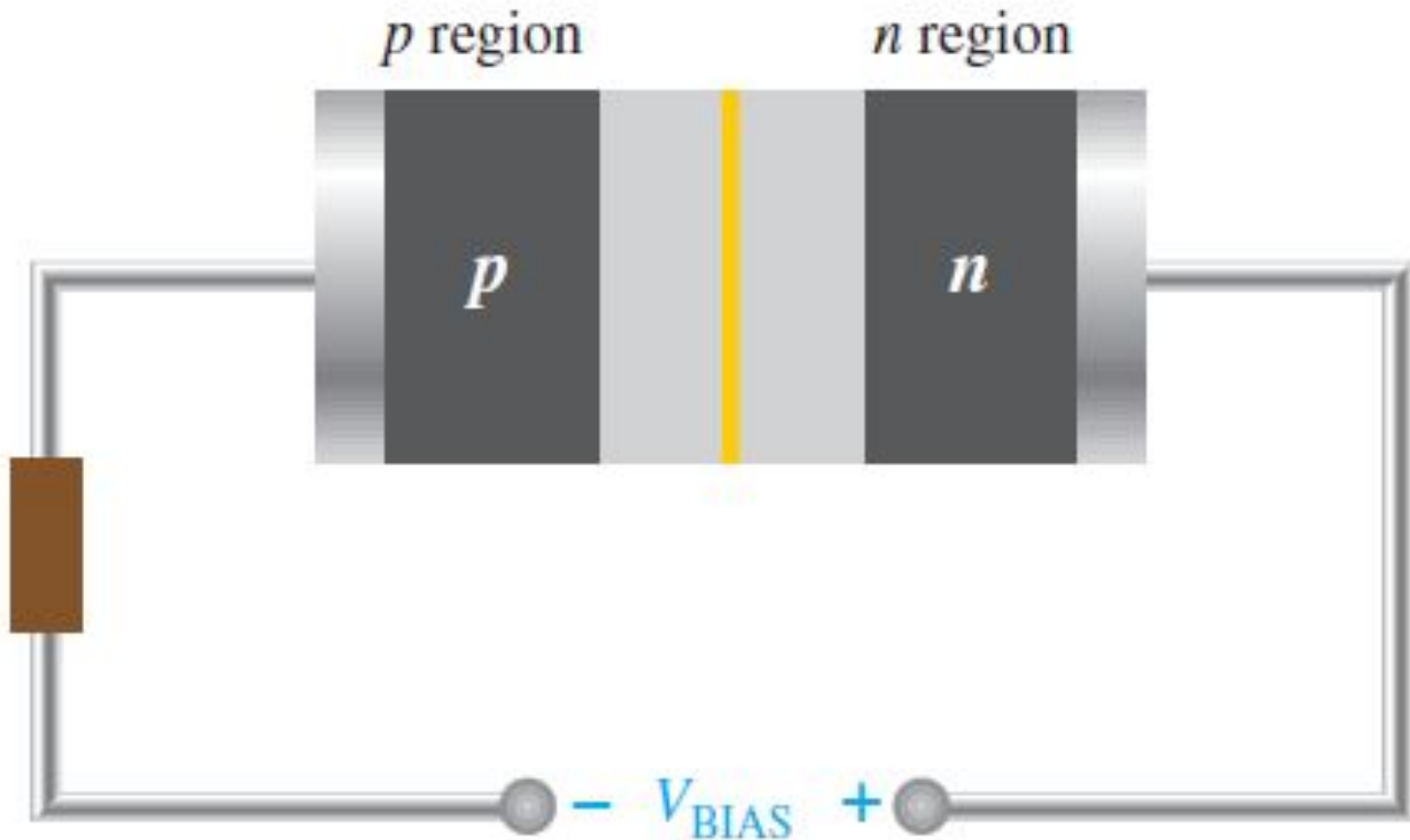
(b) At equilibrium



Reverse Bias

- **Reverse bias** is the condition that essentially prevents current through the diode.
- Figure shows a dc voltage source connected across a diode in the direction to produce reverse bias. This external bias voltage is designated as V_{BIAS} just as it was for forward bias.
- Notice that the positive side of V_{BIAS} is connected to the n region of the diode and the negative side is connected to the p region.
- Also note that the depletion region is shown

Reverse Bias (Contd.)





Reverse Bias

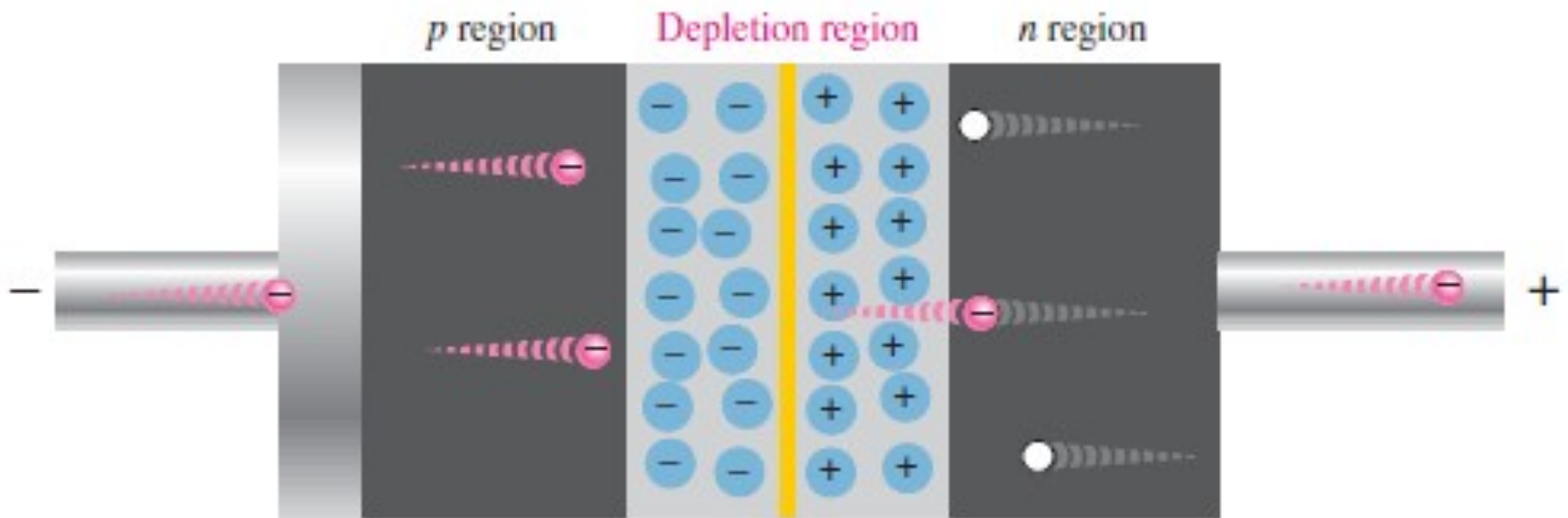
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- In the p region, electrons from the negative side of the voltage source enter as valence electrons and move from hole to hole toward the depletion region where they create ions. This results in a negative depletion region and a depletion of the majority carriers. The flow of valence electrons can be viewed as holes being “pulled” toward the positive side.

Reverse Bias (Contd.)

- The initial flow of charge carriers is transitional and lasts for only a very short time after the reverse-bias voltage is applied. As the depletion region widens, the availability of majority carriers decreases. As more of the n and p regions become depleted of majority carriers, the electric field between the positive and negative ions increases in strength until the potential across the depletion region equals the bias voltage. At this point, the net flow of charge carriers ceases except for a very small reverse current that can usually be neglected.

Reverse Bias (Contd.)





Reverse Current

- The extremely small current that exists in reverse bias after the transition current dies out is caused by the minority carriers in the n and p regions that are produced by thermally generated electron-hole pairs. The small number of free minority electrons in the p region are “pushed” toward the pn junction by the negative bias voltage. When these electrons reach the wide depletion region, they “fall down the energy hill” and combine with the minority holes in the n region as valence electrons and flow toward the positive bias voltage, creating a small hole current.



Reverse Breakdown

- Normally, the reverse current is so small that it can be neglected. However, if the external reverse-bias voltage is increased to a value called the *breakdown voltage*, the reverse current will drastically increase.
- The high reverse-bias voltage imparts energy to the free minority electrons so that as they speed through the p region, they collide with atoms with enough energy to knock valence electrons out of orbit and into the conduction band.
- The newly created conduction electrons are also high in energy and repeat the process. If one electron knocks only two others out of their valence orbit during its travel through the p region, the numbers quickly multiply.



Reverse Breakdown (Contd.)

- As these high-energy electrons go through the depletion region, they have enough energy to go through the n region as conduction electrons, rather than combining with holes.
- The multiplication of conduction electrons just discussed is known as the **avalanche effect**, and reverse current can increase dramatically if steps are not taken to limit the current.
- When the reverse current is not limited, the resulting heating will permanently damage the diode.
- Most diodes are not operated in reverse breakdown, but if the current is limited (by adding a series-limiting resistor for example), there is no permanent damage to the diode.



V-I Characteristic for Forward Bias

- When a forward-bias voltage is applied across a diode, there is current. This current is called the *forward current* and is designated I_F .
- Figure illustrates what happens as the forward-bias voltage is increased positively from 0 V.
- The resistor is used to limit the forward current to a value that will not overheat the diode and cause damage.
- With 0 V across the diode, there is no forward current. As you gradually increase the forward-bias voltage, the forward current *and* the voltage across the diode gradually increase, as shown in Figure.

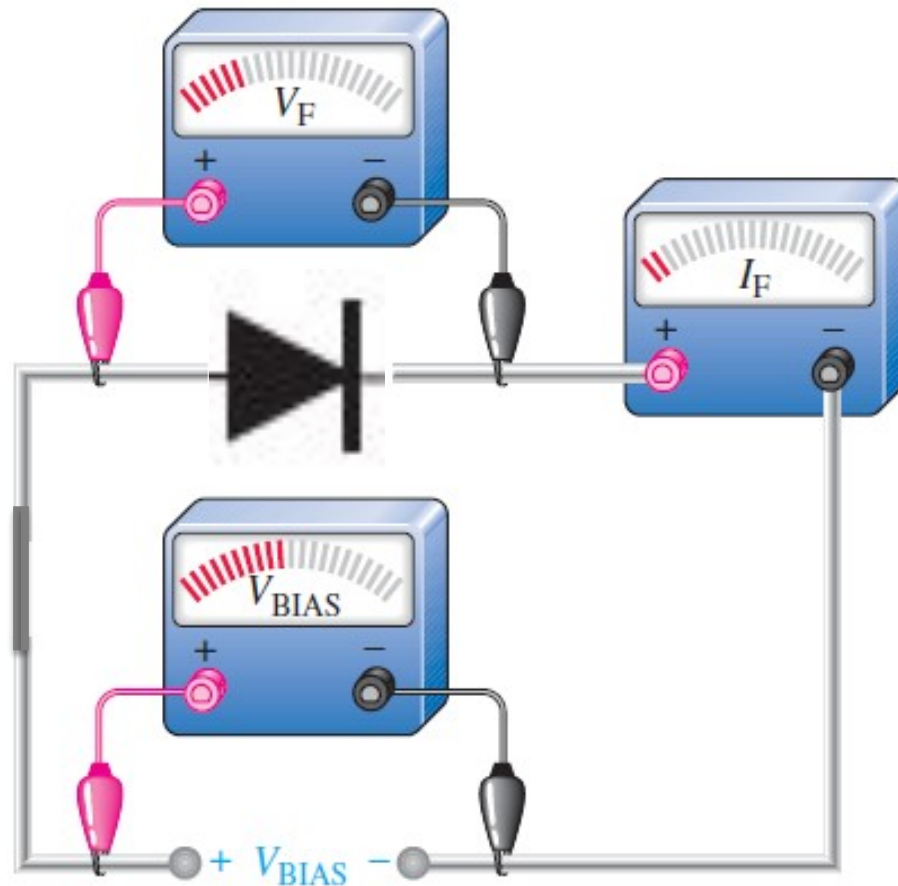
V-I Characteristic for Forward Bias (Contd.)

- A portion of the forward-bias voltage is dropped across the limiting resistor.
- When the forward-bias voltage is increased to a value where the voltage across the diode reaches approximately 0.7 V (barrier potential), the forward current begins to increase rapidly, as illustrated in Figure.
- As you continue to increase the forward-bias voltage, the current continues to increase very rapidly, but the voltage across the diode increases only gradually above 0.7 V.
- This small increase in the diode voltage above the barrier potential is due to the voltage drop across the internal dynamic resistance of the semiconductive material.

V-I Characteristic for Forward Bias (Contd.)

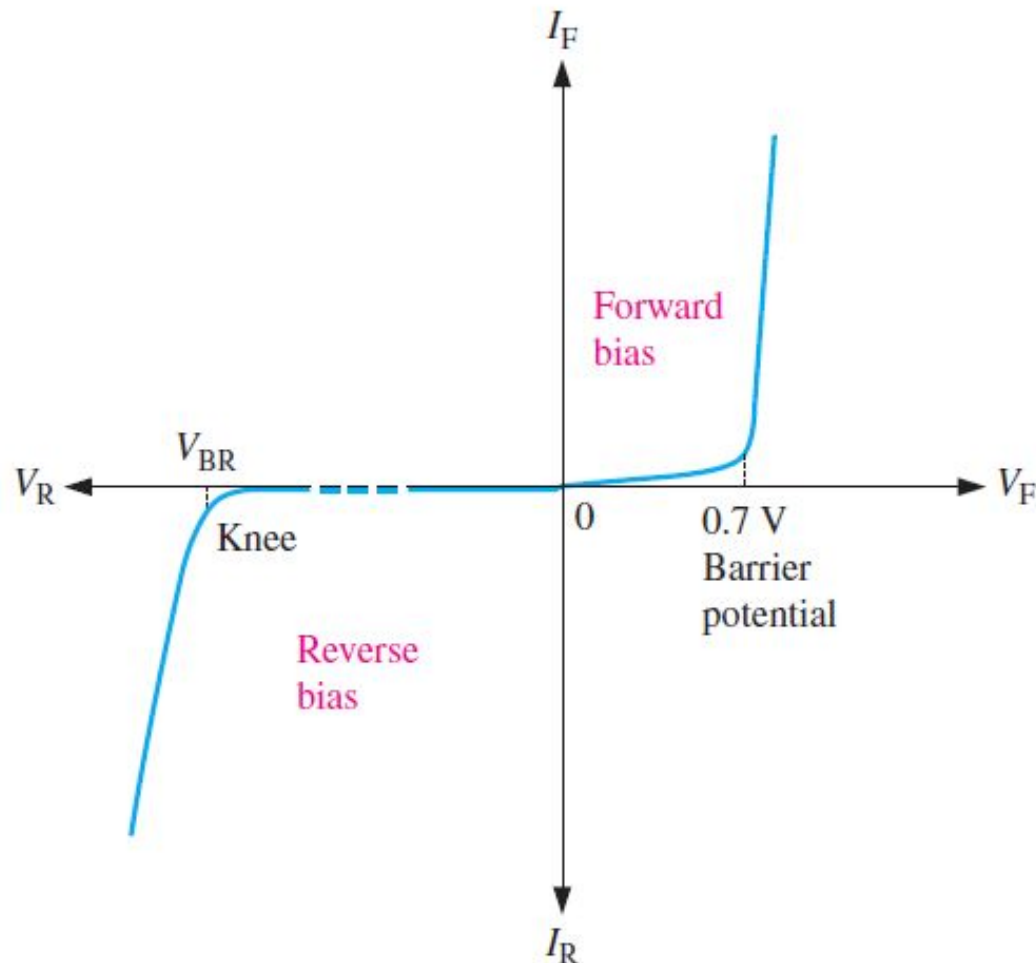
- **Graphing the V-I Curve** If you plot the results of the type of measurements shown in Figure on a graph, you get the **V-I characteristic** curve for a forward-biased diode, as shown in Figure. The diode forward voltage (V_F) increases to the right along the horizontal axis, and the forward current (I_F) increases upward along the vertical axis.

V-I Characteristic for Forward Bias (Contd.)

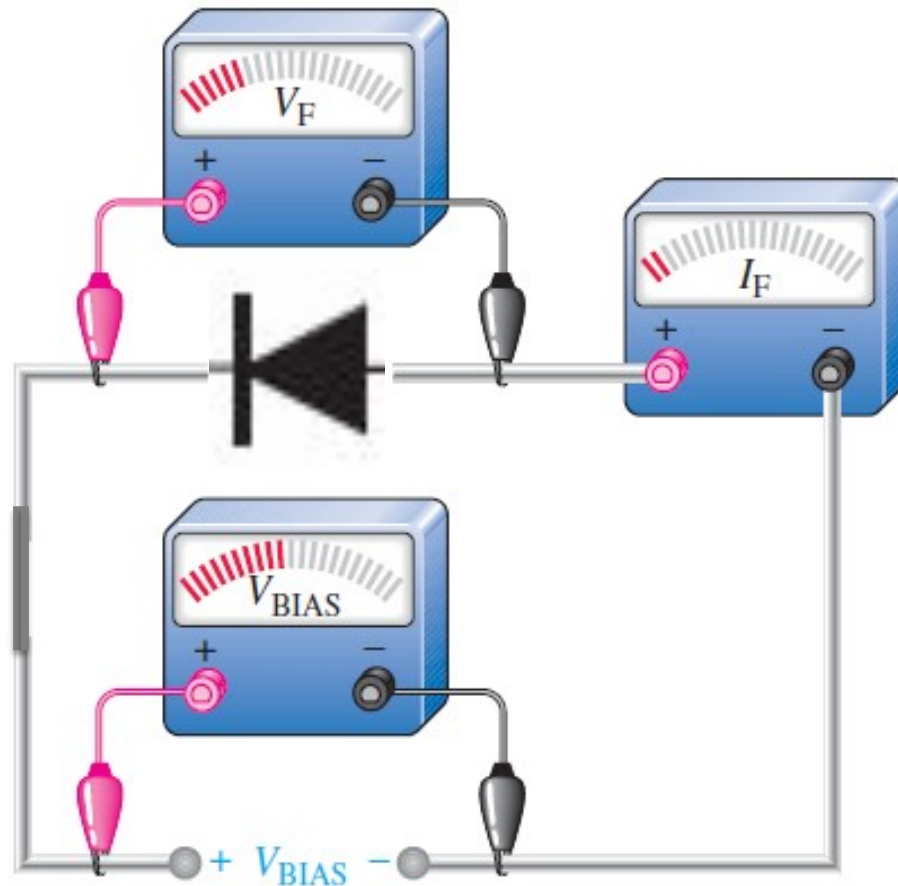


(a) Small forward-bias voltage ($V_F < 0.7$ V), very small forward current.

V-I Characteristic for Forward Bias (Contd.)

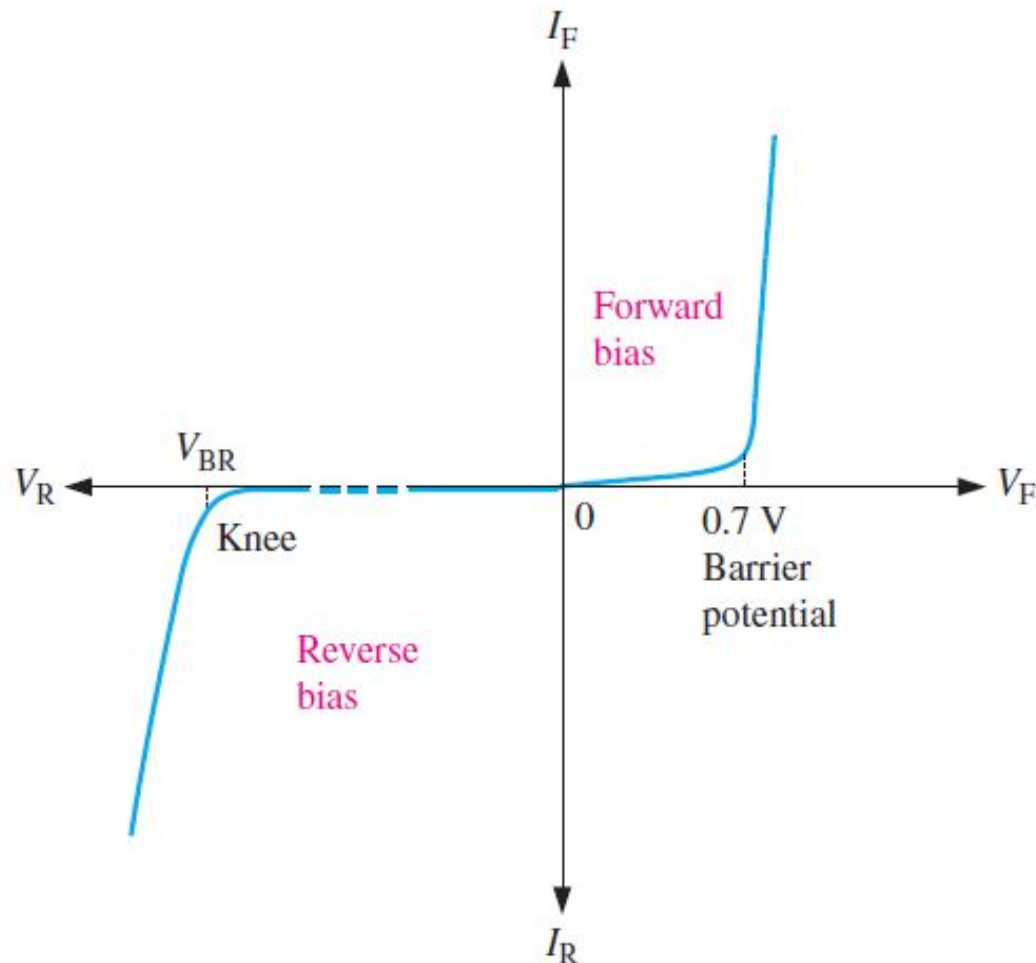


V-I Characteristic for Reverse Bias



(a) Small forward-bias voltage ($V_F < 0.7$ V), very small forward current.

V-I Characteristic for Reverse Bias (Contd.)





Diode Approximations

- The Ideal Diode Model
- The Practical Diode Model
- The Complete Diode Model



The Ideal Diode Model

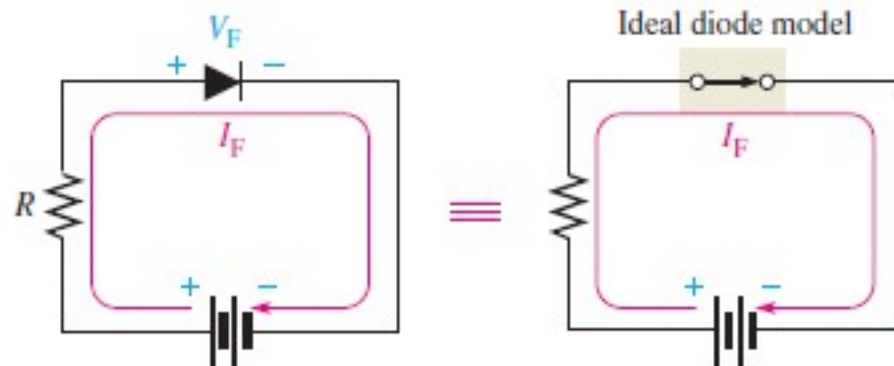
- The ideal model of a diode is the least accurate approximation and can be represented by a simple switch.
- When the diode is forward-biased, it ideally acts like a closed (on) switch, as shown in Figure (a).
- When the diode is reverse-biased, it ideally acts like an open (off) switch, as shown in part (b).
- Although the barrier potential, the forward dynamic resistance, and the reverse current

The Ideal Diode Model

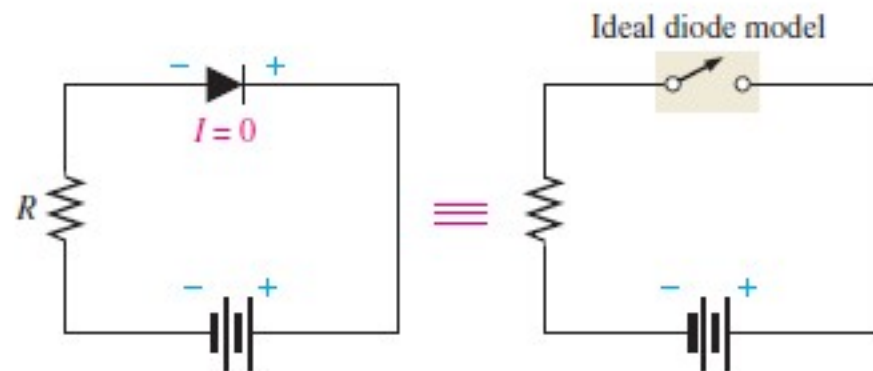
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- This model is adequate for most troubleshooting when you are trying to determine if the diode is working properly.
- In Figure (c), the ideal $V-I$ characteristic curve graphically depicts the ideal diode operation.
- Since the barrier potential and the forward dynamic resistance are neglected, the diode is assumed to have a zero voltage across it when forward-biased, as indicated by the portion of the curve on the positive vertical axis.

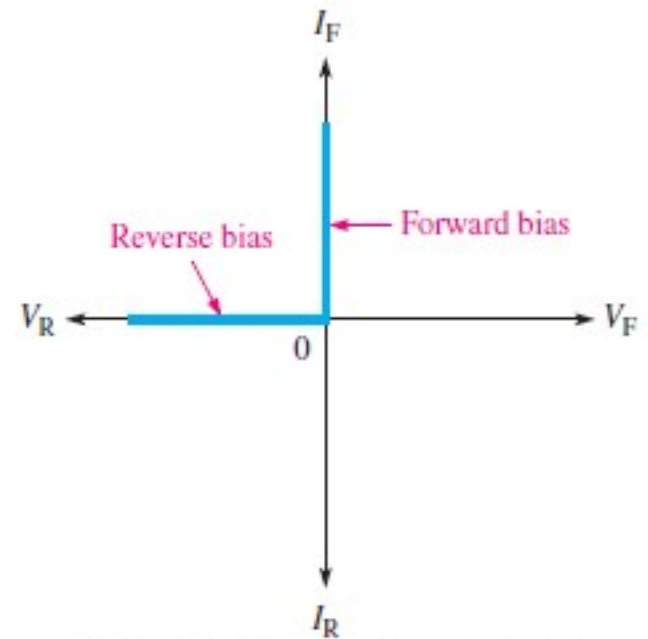
The Ideal Diode Model



(a) Forward bias



(b) Reverse bias



(c) Ideal V-I characteristic curve (blue)

The Ideal Diode Model (Contd.)

$$V_F = 0 \text{ V}$$

- The forward current is determined by the bias voltage and the limiting resistor using Ohm's law.

$$I_F = \frac{V_{BIAS}}{R_{LIMIT}}$$

- Since the reverse current is neglected, its value is assumed to be zero, as indicated in figure (c) by the portion of the curve on the negative horizontal axis.

$$I_R = 0 \text{ A}$$

- The reverse voltage equals the bias voltage.

$$V_R = V_{BIAS}$$

The Practical Diode Model

- The practical model includes the barrier potential. When the diode is forward-biased, it is equivalent to a closed switch in series with a small equivalent voltage source (V_F) equal to the barrier potential (0.7 V) with the positive side toward the anode, as indicated in Figure (a).
- This equivalent voltage source represents the barrier potential that must be exceeded by the bias voltage before the diode will conduct and is not an active source of voltage.
- When conducting, a voltage drop of 0.7 V appears across the diode. When the diode is reverse-biased, it is equivalent to an open switch just as in the ideal model, as shown in Figure (b).
- The barrier potential does not affect reverse bias, so it is not a factor.
- The characteristic curve for the practical diode model is shown in Figure (c). Since the barrier potential is included and the dynamic resistance is neglected, the diode is assumed to have a voltage across it when forward-biased, as indicated by the portion of the curve to the right of the origin.

The Practical Diode Model (Contd.)

$$V_F = 0.7 \text{ V}$$

- The forward current is determined as follows by first applying Kirchhoff's voltage

law to circuit:

$$V_{BIAS} - V_F - I_F R_{LIMIT} = 0$$

- Substituting and solving for I_F ,

$$I_F = \frac{V_{BIAS} - V_F}{R_{LIMIT}}$$

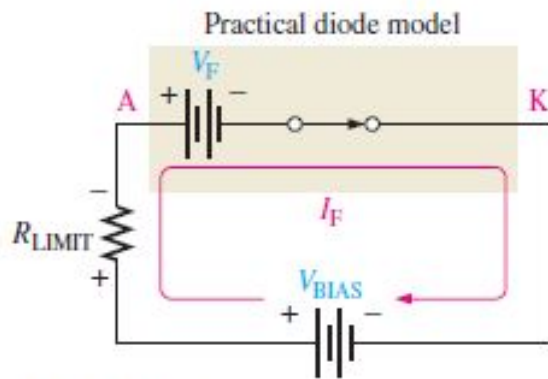
- The diode is assumed to have zero reverse current, as indicated by the portion of the curve on the negative horizontal axis.

$$I_R = 0 \text{ A}$$

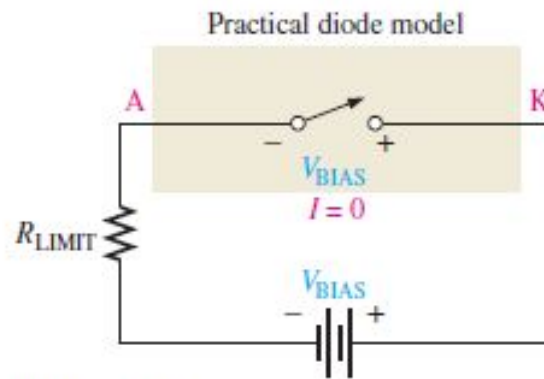
$$V_R = V_{BIAS}$$

- The practical model is useful when you are troubleshooting in lower-voltage circuits. In these cases, the 0.7 V drop across the diode may be significant and should be taken into account. The practical model is also useful when you are designing basic diode circuits.

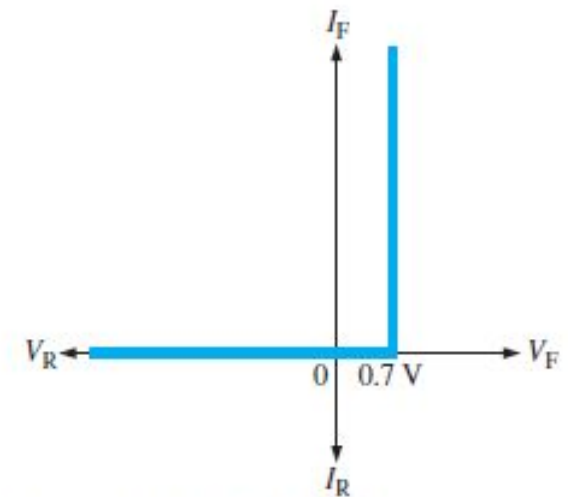
The Practical Diode Model (Contd.)



(a) Forward bias



(b) Reverse bias



(c) Characteristic curve (silicon)



The Complete Diode Model

- The complete model of a diode is the most accurate approximation and includes the barrier potential, the small forward dynamic resistance and the large internal reverse resistance. The reverse resistance is taken into account because it provides a path for the reverse current, which is included in this diode model.
- When the diode is forward-biased, it acts as a closed switch in series with the equivalent barrier potential voltage (V_B) and the small forward dynamic resistance as indicated in Figure (a).
- When the diode is reverse-biased, it acts as an open switch in parallel with the large internal reverse resistance as shown in Figure (b). The barrier potential does not affect reverse bias, so it is not a factor.
- The characteristic curve for the complete diode model is shown in Figure (c). Since the barrier potential and the forward dynamic resistance are included, the diode is assumed to have a voltage across it when forward-biased.
- This voltage (V_F) consists of the barrier potential voltage plus the small voltage drop across the dynamic resistance, as indicated by the portion of the curve to the right of the origin.

The Complete Diode Model (Contd.)

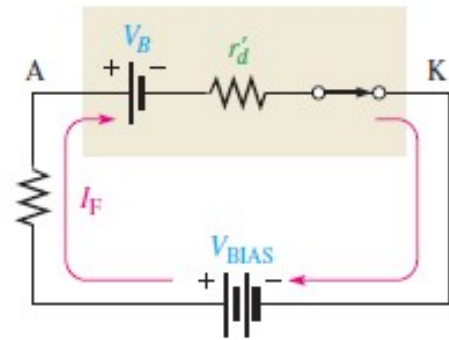
- The curve slopes because the voltage drop due to the dynamic resistance increases as the current increases. For the complete model of a silicon diode, the following formulas apply:

$$V_F = V_{BIAS} = 0.7 \text{ V} + I_F r'_d$$

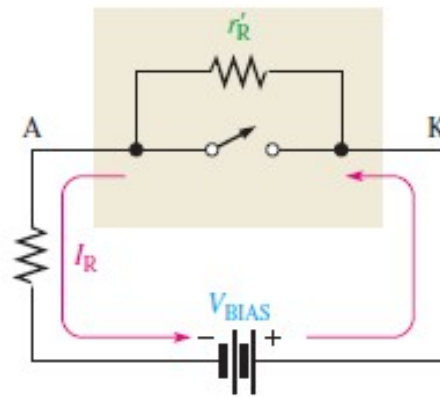
$$I_F = \frac{V_{BIAS} - 0.7 \text{ V}}{R_{LIMIT} + r'_d}$$

- The reverse current is taken into account with the parallel resistance and is indicated by the portion of the curve to the left of the origin. The breakdown portion of the curve is not shown because breakdown is not a normal mode of operation for most diodes.
- For troubleshooting work, it is unnecessary to use the complete model, as it involves complicated calculations. This model is generally suited to design problems using a computer for simulation. The ideal and practical models are used for circuits in this text, except in the following example, which illustrates the differences in the three models.

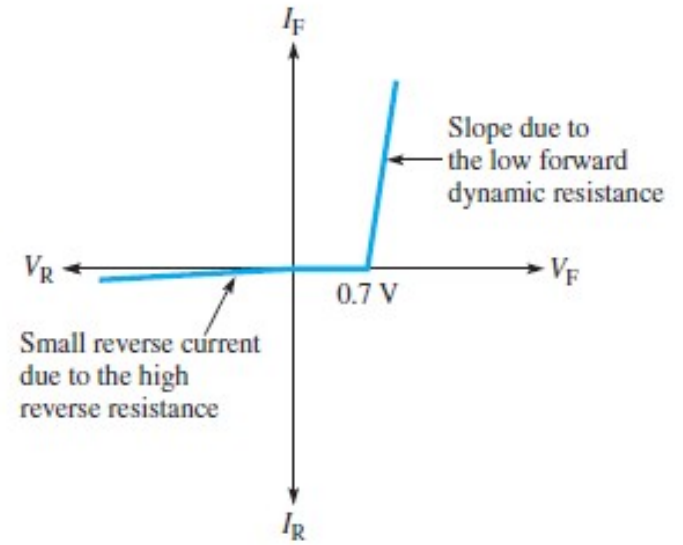
The Complete Diode Model (Contd.)



(a) Forward bias



(b) Reverse bias



(c) V-I characteristic curve

Diode (Example)

- Determine the forward voltage and forward current for the diode in Figure (a) for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume $r'_d = 10 \Omega$ at the determined value of forward current.
- Determine the reverse voltage and reverse current for the diode in Figure (b) for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume $I_R = 1 \mu A$.

Diode (Example Contd.)

